PUSHFORWARD OF THE DIVISOR OF A PULLED BACK SECTION EQUALS THE DEGREE TIMES THE DIVISOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The norm compatibility of the pushforward of a principal divisor. Same chapter chow.tex, label lemma-proper-pushforward-alteration (proof lines 4613 and 4624, claimed as S14-algebra-pushforward-alteration). Gives the equality of cycles for the divisor of a rational function on an integral scheme, from which the statement follows after reducing to the case of the structure sheaf.

Source reference. Original lemma 02RT: sources/stacks-chow/pinned-original-source.tex, source lines 2645–2645. Navigation pointer for \ref{lemma-proper-pushforward-alteration} used inside the counted span (line 4613); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 4591); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X, Y be locally of finite type over S. Let $f : X \to Y$ be a proper morphism. Let $\mathcal{L}$ be an invertible sheaf on Y. Let s be a nonzero meromorphic section s of $\mathcal{L}$ on Y. Assume X, Y integral, f dominant, and $\dim_\delta(X) = \dim_\delta(Y)$. Then*

$$f_* \left(div_{f^*\mathcal{L}}(f^* s) \right) = [R(X) : R(Y)] div_{\mathcal{L}}(s).$$

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as cycles on Y. In particular

$$f_*(c_1(f^*\mathcal{L}) \cap [X]) = [R(X) : R(Y)]c_1(\mathcal{L}) \cap [Y] = c_1(\mathcal{L}) \cap f_*[X]$$

Proof. The last equation follows from the first since $f_*[X] = [R(X) : R(Y)][Y]$ by definition. It turns out that we can re-use Lemma 02RT to prove this. Namely, since we are trying to prove an equality of cycles, we may work locally on Y. Hence we may assume that $\mathcal{L} = \mathcal{O}_Y$. In this case s corresponds to a rational function $g \in R(Y)$, and we are simply trying to prove

$$f_* \left(\mathrm{div}_X(g)\right) = [R(X) : R(Y)]\mathrm{div}_Y(g).$$

Comparing with the result of the aforementioned Lemma 02RT we see this true since $\mathrm{Nm}_{R(X)/R(Y)}(g) = g^{[R(X):R(Y)]}$ as $g \in R(Y)^*$. □